App Dev Project Report

1. Student Details

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About Me

Hello, I am Piyush Kumar. I am a student of the IIT Madras BS Degree Program in Data Science and Applications. I am passionate about software development, data science, and building practical applications that solve real-world problems.

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2. Project Details

Project Title

→ ***Trekking Management Application***

Problem Statement

Adventure and trekking organizations often rely on spreadsheets, phone calls, and manual coordination to manage trekking events. Such approaches make it difficult to manage trek approvals, assign staff, prevent overbooking, track participant registrations, and maintain trekking history.

The objective of this project is to design and build a web-based Trekking Management Application that enables administrators, trek staff, and trekkers to efficiently manage trekking activities through a centralized platform.

Approach

The application was developed using Flask with a modular architecture consisting of separate models, routes, templates, and configuration files.

The application implements role-based access control, booking management, trek assignment workflows, and booking history tracking while ensuring that users cannot book closed treks or exceed available capacity.

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3. AI / LLM Declaration

I used ChatGPT (GPT-5) as an assistant during the development process.

The usage was limited to:

- Bootstrap styling suggestions and UI improvements
- Guidance on Flask and SQLAlchemy configurations
- Explanations of framework features and implementation approaches
- Code review and best-practice recommendations

The estimated extent of AI/LLM usage is approximately **15%** of the overall project effort.

All application design decisions, database modeling, route implementation, debugging, integration, testing, and final code modifications were completed manually.

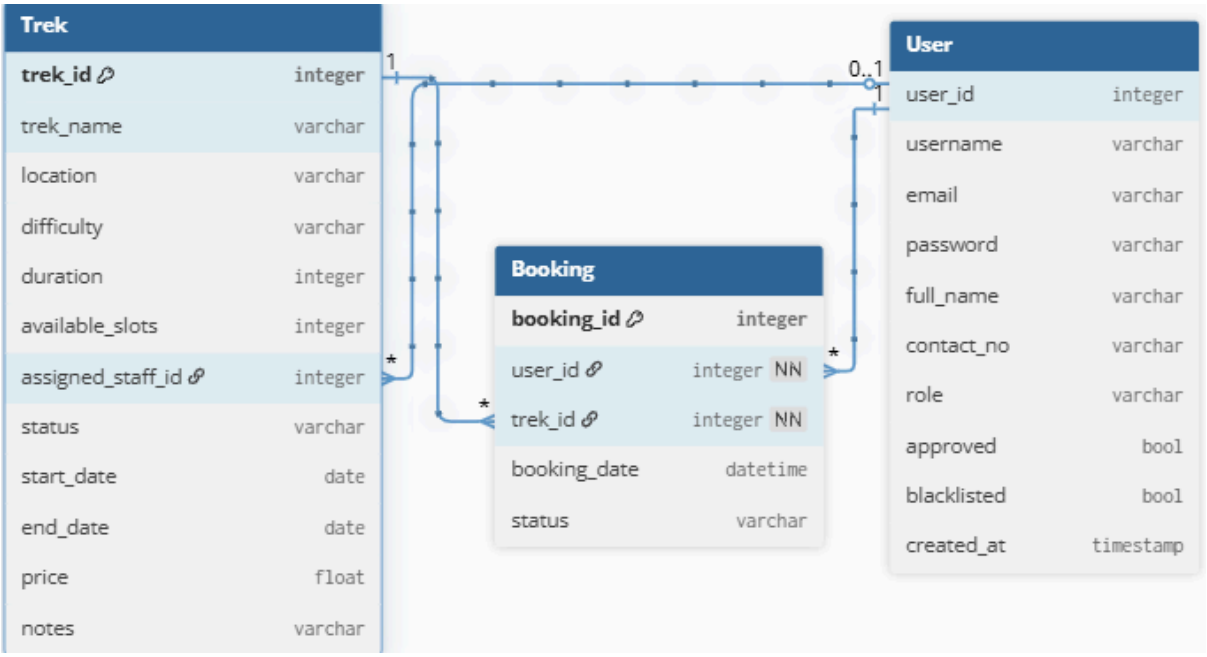
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4. Technologies and Frameworks

Technology / Library	**Purpose**
Flask	Backend web framework
SQLAlchemy	ORM for database operations
SQLite	Application database
Jinja2	Dynamic HTML template rendering
Bootstrap 5	Responsive frontend styling
Flask-Login	User authentication and session management
Werkzeug Security	Password hashing and verification
HTML5	Frontend structure
CSS3	Additional styling and customization

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5. Database Schema / ER Diagram



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6. Implemented Features

Authentication

- User registration, Staff registration, Login and logout functionality, Secure password hashing, Session management using Flask-Login

Admin Features

- Create treks, Edit treks, Delete treks, Approve staff registrations, Blacklist users and staff, Assign staff to treks, View all users, View all staff, View all bookings, Search users, staff, and treks.

Trek Staff Features

- View assigned treks, Update trek status, Update available slots, View participant lists, Start treks, Mark treks as completed

User Features

- View available treks, Search treks, Filter treks by difficulty and location, Book treks, View booking status, View trekking history, Edit profile

Booking Management

- Prevent overbooking, Prevent duplicate bookings, Allow booking only for open treks, Maintain complete booking history

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7. Video Presentation

Drive Link

https://drive.google.com/file/d/1UTv91B-dOU-1nd0XfnTOsupE2ekL5ey/view?usp=sharing